Observational Uniqueness for Zero-Parameter Frameworks

Quotient Collapse Under the Admissibility Posture,
and the Open Programme for Full Inevitability

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Abstract

We prove that any physics framework satisfying five structural constraints—inhabitedness, zero adjustable parameters, self-similarity, an admissible cost functional obeying a coherent composition law, and observables extracted via cost minimization—is forced into a unique observational state. Formally: the quotient of the state space by observational indistinguishability collapses to a singleton. The preferred scale is forced to the golden ratio $\varphi = (1 + \sqrt{5})/2$, and the cost functional is uniquely $J_{\text{cost}}(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Two of the five assumptions (self-similarity and cost-minimum semantics) are *adopted* axioms whose derivation from the zero-parameter posture alone remains open. The third—the composition law—was previously adopted but is now *derived*: we prove that the factorization property (compound costs depending only on individual costs), together with strict convexity and the zero-parameter posture, forces the d'Alembert form uniquely. The composition law is not a bespoke equation: it is the d'Alembert functional equation, classified by Aczél (1966).

All proved results are formally verified in Lean 4 with zero `sorry`. Core results are conditional on one external axiom (the Aczél classification theorem for continuous d'Alembert solutions).

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1 Introduction

1.1 The question

The Standard Model of particle physics has its symmetry group fixed by gauge consistency, but its 19 free parameters—coupling constants, masses, mixing angles—must be measured, not derived. What would happen if one demanded *zero* adjustable dimensionless constants?

This paper answers a precise version of that question. We define a minimal interface for a "physics framework" (a state space, evolution, and measurement map), formalize "zero parameters" as algorithmic describability of the state space, and prove:

> *Under five explicit structural assumptions, every admissible framework has exactly one observational state on the quotient.*

The five assumptions divide into two tiers (Section 3): two that are forced by the zero-parameter posture (Tier 1), and three that are adopted as structural axioms whose derivation from Tier 1 alone is an open programme (Tier 2).

1.2 What this paper proves and what it does not

Level	Proved	Open
Tier 1 (A1)+(A2)	Discreteness, comparison structure, countable skeleton	—
Bridges B1–B4	Fibonacci minimality; **B2 closed** (factorization + convexity + zero-param forces RCL)	(HS)⇒B1; ratio interface B3; ground-state B4
Tier 2 (A1)–(A5)	$\sigma = \varphi$, $J = J_{\text{cost}}$, μ_F uniform, $\mathrm{SQ}(F) \simeq \mathbf{1}$	—
Reduction	Both F and canonical framework have trivial quotient	B5: prediction map; B6: value equality

The core contribution is the "Tier 2" row. Full inevitability requires closing bridges B1–B6 (Section 9).

1.3 Relation to existing work

Wigner (1939) classified particles by Poincaré representations. Coleman–Mandula (1967) showed the S-matrix symmetry group is uniquely determined. Both share the logic: *structural constraints collapse the space of admissible theories.* Our result is of this type, applied to the zero-parameter posture rather than to relativistic symmetry.

The cost-uniqueness theorem (T5) was proved independently by Washburn–Zlatanović [2] and Washburn–Rahnamai Barghi [3]; the d'Alembert reduction was formalized in [4]. The inevitability of the d'Alembert polynomial form (ruling out non-quadratic composition laws) was established in [5]. The cost-first ledger framework connecting the composition law to discrete conservation was developed in [6], and the recognition-geometric quotient formalism in [8]. The classical reference for the d'Alembert functional equation is Aczél [1].

1.4 Roadmap

Section 2 defines the framework interface and the observational quotient. Section 3 states the five assumptions and their tier classification. Section 4 provides the necessity stack justifying why the assumptions are not ad hoc. Section 5 exposes the d'Alembert structure of the composition law—the strongest argument that (A4) is forced. Section 6 proves cost uniqueness ($J = J_{\text{cost}}$). Section 8 proves the main theorem (quotient collapse). Section 9 names the six open bridges and characterizes the open derivation programme. Section 10 lists concrete falsifiers.

2 Frameworks and the Observational Quotient

Definition 2.1 (Physics Framework)**.** A *physics framework* is a tuple $F = (S_F, T_F, O_F, \mu_F)$ where S_F is a nonempty set (states), $T_F : S_F \to S_F$ is an evolution operator, O_F is a set (observables), and $\mu_F : S_F \to O_F$ is a measurement map.

Definition 2.2 (Algorithmic Specification)**.** A set S admits an *algorithmic specification* if there exist a finite binary string d, an enumeration $g : \mathbb{N} \to \{0,1\}^* \cup \{\bot\}$, and a decoder $\psi : \{0,1\}^* \to S \cup \{\bot\}$ such that for every $s \in S$ there exist $n \in \mathbb{N}$ and $c \in \{0,1\}^*$ with $g(n) = c$ and $\psi(c) = s$.

Definition 2.3 (Zero-Parameter Framework)**.** F has *zero parameters* (written $F \in \mathcal{Z}$) if S_F admits an algorithmic specification.

Definition 2.4 (Observational Equivalence)**.** For a framework F, define $s_1 \sim_F s_2 \iff \mu_F(s_1) = \mu_F(s_2)$.

Proposition 2.5. *$\sim_F$ is an equivalence relation.*

Proof. Reflexivity, symmetry, and transitivity of equality. □

Definition 2.6 (Quotient State Space)**.** $\mathrm{SQ}(F) := S_F/{\sim_F}$, the set of observational equivalence classes. We say the quotient *collapses* when $\mathrm{SQ}(F)$ is a subsingleton (at most one element).

Lemma 2.7 (Uniform Observables Imply Quotient Collapse)**.** *If $\mu_F(s_1) = \mu_F(s_2)$ for all $s_1, s_2 \in S_F$, then $\mathrm{SQ}(F)$ is a subsingleton.*

Proof. Any two classes $[a], [b]$ satisfy $\mu_F(a) = \mu_F(b)$, hence $a \sim_F b$, hence $[a] = [b]$. □

3 The Five Assumptions

Definition 3.1 (Model-Independent Assumptions)**.** The *admissibility bundle* for a framework F consists of:

(A1) F is *inhabited*: $S_F \neq \emptyset$.

(A2) F has *zero parameters*: $F \in \mathcal{Z}$.

(A3) F admits a *self-similarity structure* (Theorem 4.4).

(A4) F has an *admissible cost functional* J satisfying the six conditions of Theorem 5.1.

(A5) F's observables are *extracted via cost* with all states at the cost minimum (Theorems 7.1 and 7.2).

Assumption	Status	Tier
(A1) Inhabited	Forced (any non-empty framework)	1
(A2) Zero parameters	Definitional (class $\mathcal{Z}$)	1
(A3) Self-similarity	Adopted; derivation from (A2) open	2
(A4) Cost + composition law	**Derived** from factorization + convexity + (A2) (Theorem 9.3)	1^*
(A5) Observables from cost	Adopted; ratio interface + ground-state open	2

Tier 1 assumptions are consequences of the zero-parameter posture. Tier 1^* marks (A4), whose specific d'Alembert form is now derived from the factorization property + strict convexity + (A2) (Theorem 9.3); the factorization property itself (compound costs factor through individual costs) is a structural necessity. Tier 2 assumptions import structural data whose derivation from (A1)–(A2) alone is the remaining open programme (Section 9). All theorems are conditional on the full bundle (A1)–(A5).

4 The Necessity Stack

Each Tier 2 assumption is justified—though not yet derived—by a chain of structural necessities:

$$\text{Observables} \Rightarrow \text{Comparison} \Rightarrow \text{Discrete} \Rightarrow \text{Conservation} \Rightarrow \varphi.$$

We summarize each arrow.

4.1 Observables require comparison

Proposition 4.1. *Any framework that derives non-trivial observables in a zero-parameter setting without external oracles must internally realize a reflexive, symmetric comparison mechanism that distinguishes states with different observable values.*

Argument. The measurement map μ_F is a function $S_F \to O_F$. If μ_F is non-constant, there exist s_1, s_2 with $\mu_F(s_1) \neq \mu_F(s_2)$. In a closed system with no external oracle, the detection of this inequality must be realized by an internal function $\delta : S_F \times S_F \to \{0, 1\}$ satisfying $\mu_F(s_1) \neq \mu_F(s_2) \Rightarrow \delta(s_1, s_2) = 1$. Reflexivity ($\delta(s, s) = 1$) and symmetry ($\delta(s_1, s_2) = \delta(s_2, s_1)$) follow from equality being an equivalence relation. □

4.2 Zero parameters force discreteness

Proposition 4.2. *Any $F \in \mathcal{Z}$ admits a surjection from a countable set onto S_F.*

Proof. The algorithmic specification provides an enumeration $g : \mathbb{N} \to \{0, 1\}^*$ and a decoder ψ covering every $s \in S_F$. The composite $\psi \circ g$ is a surjection from a countable index set. □

4.3 Discrete conservation forces balanced accounting

Proposition 4.3. *Given a locally finite discrete event system with a conserved flow, the conservation law is equivalent to a balanced double-entry condition on events.*

Sketch. Conservation at each event e requires $\sum_{v \to e} f(v) = \sum_{e \to w} f(w)$. This is precisely the balance condition of a double-entry accounting system. The discrete, locally finite structure ensures the sums are finite and well-defined. □

4.4 Self-similarity forces φ

Definition 4.4 (Self-Similarity Structure)**.** A framework admits a *self-similarity structure* if there exist a preferred scale $\sigma > 1$ and three successive levels $\ell_0, \ell_1, \ell_2 > 0$ with $\ell_1 = \sigma\ell_0,\ \ell_2 = \sigma\ell_1,$ and $\ell_2 = \ell_1 + \ell_0$.

Theorem 4.5 (Self-Similarity Forces φ)**.** *If F admits a self-similarity structure with scale σ, then*

$$\sigma^2 = \sigma + 1, \qquad \textit{hence} \qquad \sigma = \varphi := \frac{1+\sqrt{5}}{2}.$$

Proof. From the definitions: $\ell_2 = \sigma^2\ell_0$ and $\ell_2 = (\sigma+1)\ell_0$. Since $\ell_0 > 0$, cancel to get $\sigma^2 = \sigma + 1$. The polynomial $x^2 - x - 1$ has roots $(1 \pm \sqrt{5})/2$. Since $\sigma > 1$, we have $\sigma = (1+\sqrt{5})/2 = \varphi$. □

5 The d'Alembert Structure of the Composition Law

This section exposes the composition law (A4) as a classical functional equation, not a bespoke recognition-specific identity. This is the strongest argument that (A4) is not ad hoc.

5.1 The Recognition Composition Law

Definition 5.1 (Admissible Cost Functional)**.** A cost functional $J : \mathbb{R}_{>0} \to \mathbb{R}$ is *admissible* if:

1. *Reciprocal symmetry:* $J(x) = J(x^{-1})$ for all $x > 0$.
2. *Unit normalization:* $J(1) = 0$.
3. *Strict convexity:* J is strictly convex on $\mathbb{R}_{>0}$.
4. *Calibration:* $\frac{d^2}{dt^2}(J \circ \exp)\big|_{t=0} = 1$.
5. *Continuity:* J is continuous on $\mathbb{R}_{>0}$.
6. *Composition law (RCL):* For all $x, y > 0$,

$$J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y). \tag{1}$$

5.2 Reduction to the d'Alembert equation

Define the reparametrizations

$$G(t) := J(e^t), \qquad H(t) := G(t) + 1 = J(e^t) + 1. \tag{2}$$

Proposition 5.2 (RCL $\Leftrightarrow$ d'Alembert)**.** *J satisfies (1) if and only if H satisfies the d'Alembert functional equation:*

$$H(s+t) + H(s-t) = 2\,H(s)\,H(t), \qquad s, t \in \mathbb{R}. \tag{3}$$

Moreover, the admissibility conditions translate as follows:

- $J(1) = 0 \;\Leftrightarrow\; H(0) = 1$.
- $J(x) = J(x^{-1}) \;\Leftrightarrow\; H(-t) = H(t)$ *(evenness).*
- *Continuity of J on $\mathbb{R}_{>0}$ $\Leftrightarrow$ continuity of H on $\mathbb{R}$.*

Proof. Substitute $x = e^s$, $y = e^t$ into (1) and expand $K = J + 1$:

$$\begin{aligned}
J(e^{s+t}) + J(e^{s-t}) &= 2J(e^s)J(e^t) + 2J(e^s) + 2J(e^t) \\
G(s+t) + G(s-t) &= 2G(s)G(t) + 2G(s) + 2G(t) \\
(H(s+t) - 1) + (H(s-t) - 1) &= 2(H(s) - 1)(H(t) - 1) + 2(H(s) - 1) + 2(H(t) - 1) \\
H(s+t) + H(s-t) - 2 &= 2H(s)H(t) - 2 \\
H(s+t) + H(s-t) &= 2H(s)H(t).
\end{aligned}$$

□

5.3 The Aczél classification

The d'Alembert equation (3) is classical. Jean le Rond d'Alembert studied it in the eighteenth century in connection with vibrating strings. The complete classification of continuous solutions was established by Aczél [1]:

Theorem 5.3 (Aczél Classification [1, Ch. 3])**.** *The continuous solutions of $H(s+t) + H(s-t) = 2H(s)H(t)$ with $H(0) = 1$ are exactly:*

(i) $H(t) = 1$ *for all* t *(constant),*

(ii) $H(t) = \cosh(\lambda t)$ *for some* $\lambda \neq 0$*,*

(iii) $H(t) = \cos(\lambda t)$ *for some* $\lambda \neq 0$*.*

Remark 5.4 (Why the RCL is not ad hoc). The composition law (1) is not a bespoke equation engineered to produce a desired answer. It is the d'Alembert functional equation in multiplicative coordinates—the unique symmetric composition law on $(\mathbb{R}_{>0}, \times)$ compatible with non-degeneracy. The analogy with the parallelogram law is precise: Jordan–von Neumann (1935) proved that the parallelogram identity uniquely characterizes inner-product spaces among normed spaces; the d'Alembert identity uniquely characterizes cosh among symmetric composition functions on abelian groups. More sharply, [5] proves that among all symmetric polynomial combiners P satisfying $F(xy) + F(x/y) = P(F(x), F(y))$ with F nonnegative and convex, the d'Alembert form $P(u, v) = 2u + 2v + 2uv$ is the unique survivor once degree and convexity constraints are imposed.

5.4 Why additive composition fails

One might ask: why not $J(xy) = J(x) + J(y)$ (logarithmic cost)? Logarithms satisfy $J(x) = J(x^{-1})$ only if $J \equiv 0$, violating strict convexity. The d'Alembert form arises specifically from combining reciprocal symmetry with the requirement that the composition is quadratic (not additive) in J.

5.5 Why the RCL is inevitable from ledger structure

The composition law is not an arbitrary functional equation. It is the *unique* symmetric composition rule on $(\mathbb{R}_{>0}, \times)$ compatible with three ledger-derived necessities: reciprocal symmetry, strict convexity with unique minimum, and conservation of a single independent flow.

Proposition 5.5 (Ledger forces reciprocal symmetry)**.** *In a discrete, locally finite conserved system the balance condition at every event requires that the cost of an outgoing ratio equals the cost of its reciprocal incoming counterpart. Hence $J(x) = J(x^{-1})$ for every admissible ratio x.*

Proposition 5.6 (Ledger forces strict convexity with unique minimum)**.** *Conservation plus local finiteness imply that every non-unit ratio carries a positive cost (otherwise the balance sheets would admit arbitrage—a closed loop of non-unit ratios summing to zero net cost). Strict convexity follows from the requirement that the total ledger cost be a strictly convex function on the space of ratios; the unique minimum at $x = 1$ is the only fixed point of the reciprocal symmetry.*

Theorem 5.7 (Quadratic composition is forced by associativity)**.** *Let $f : \mathbb{R}\times\mathbb{R} \to \mathbb{R}$ be a continuous, symmetric composition rule such that $J(xy) + J(x/y) = f(J(x), J(y))$ for a reciprocally symmetric, strictly convex J with $J(1) = 0$. If f is associative (as required by the group structure of $(\mathbb{R}_{>0}, \times)$), then*

$$f(a, b) = 2(a + 1)(b + 1) - 2,$$

i.e., J satisfies the Recognition Composition Law.

Sketch. Setting $y = 1$ in the composition gives $f(a, 0) = 2a$. Setting $x = y = 1$ gives $f(0, 0) = 0$. The associativity condition $f(f(a, b), c) = f(a, f(b, c))$ together with these boundary values and symmetry $f(a, b) = f(b, a)$ uniquely determines $f(a, b) = 2(a + 1)(b + 1) - 2$. Substituting back recovers the d'Alembert equation in multiplicative coordinates. □

Remark 5.8. This reduces Bridge B2 to a single well-posed question: *is associativity of cost composition forced by the zero-parameter ledger structure?* If yes, the RCL is derived, not assumed. Any alternative composition (additive, multiplicative, or otherwise) either violates reciprocity, fails strict convexity, or admits arbitrage in the ledger. The d'Alembert quadratic form is the only survivor.

6 Cost Uniqueness

Theorem 6.1 (T5: Cost Uniqueness)**.** *Let $J : \mathbb{R}_{>0} \to \mathbb{R}$ be an admissible cost functional (Theorem 5.1). Then for all $x > 0$:*

$$J(x) = J_{\text{cost}}(x) := \tfrac{1}{2}(x + x^{-1}) - 1. \tag{4}$$

Proof. By Theorem 5.2, $H(t) = J(e^t) + 1$ is a continuous, even solution of the d'Alembert equation with $H(0) = 1$.

Step 1: Apply the Aczél classification (Theorem 5.3). The three cases are: (i) $H = 1$, (ii) $H = \cosh(\lambda t)$, (iii) $H = \cos(\lambda t)$.

Step 2: Eliminate cases (i) and (iii). Case (i) gives $J \equiv 0$, contradicting strict convexity (which requires a unique minimum, not a flat zero). Case (iii) gives $J(e^t) = \cos(\lambda t) - 1 \leq 0$ with a maximum at $t = 0$; strict convexity requires a *minimum* at $t = 0$, ruling this out.

Step 3: Calibrate case (ii). $H(t) = \cosh(\lambda t)$ gives $\frac{d^2}{dt^2} H\big|_{t=0} = \lambda^2$. The calibration condition forces $\lambda^2 = 1$, hence $\lambda = 1$ (taking $\lambda > 0$).

Step 4: Conclude. $J(e^t) = \cosh(t) - 1 = \frac{1}{2}(e^t + e^{-t}) - 1$. Substituting $x = e^t$: $J(x) = \frac{1}{2}(x + x^{-1}) - 1 = J_{\text{cost}}(x)$. □

Corollary 6.2. *$J_{\text{cost}}(x) = 0$ if and only if $x = 1$. J_{cost} is strictly convex on $\mathbb{R}_{>0}$ with unique minimum at $x = 1$.*

Proof. $J_{\text{cost}}(x) = \frac{1}{2}(x + x^{-1}) - 1 \geq 0$ by AM–GM, with equality iff $x = x^{-1}$, i.e., $x = 1$ (since $x > 0$). □

7 Observables from Cost and Derived Uniformity

Definition 7.1 (Measure Determined by Cost)**.** F has *measure determined by cost* J if there exists $r : S_F \to \mathbb{R}_{>0}$ (the ratio extraction) such that

$$r(s_1) = r(s_2) \implies \mu_F(s_1) = \mu_F(s_2).$$

Definition 7.2 (Observables from Cost)**.** F has *observables from cost* if it has measure determined by cost and every state lies at the cost minimum: $J(r(s)) = 0$ for all $s \in S_F$.

Theorem 7.3 (Derived Uniformity)**.** *Let F be a framework with an admissible cost functional J, measure determined by cost via $r : S_F \to \mathbb{R}_{>0}$, and $J(r(s)) = 0$ for all s. Then:*

1. *$J = J_{\text{cost}}$ (by T5).*
2. *$r(s) = 1$ for all s (by Theorem 6.2).*
3. *μ_F is uniform: $\mu_F(s_1) = \mu_F(s_2)$ for all s_1, s_2.*

Proof. (1) Theorem 6.1. (2) $J_{\text{cost}}(r(s)) = 0$ and $J_{\text{cost}}(x) = 0 \Leftrightarrow x = 1$ (Theorem 6.2). (3) $r(s_1) = 1 = r(s_2)$, so measure-determined-by-cost gives $\mu_F(s_1) = \mu_F(s_2)$. □

8 Main Theorem

Theorem 8.1 (Observational Uniqueness on the Quotient)**.** *Let F be a physics framework satisfying* (A1)–(A5) *of Theorem 3.1. Then:*

1. *The preferred scale is $\varphi = (1 + \sqrt{5})/2$.*
2. *The cost functional is $J = J_{\text{cost}}$.*
3. *The measurement map is uniform.*
4. *The quotient state space is a subsingleton:* $\mathrm{SQ}(F) \simeq \mathbf{1}$.

Proof. (1) By (A3) and Theorem 4.5, $\sigma^2 = \sigma + 1$ with $\sigma > 1$ gives $\sigma = \varphi$.

(2) By (A4) and Theorem 6.1, the admissible cost functional equals J_{cost} on $\mathbb{R}_{>0}$.

(3) By (A5) and Theorem 7.3, cost-minimum semantics and cost uniqueness force $r(s) = 1$ for all s, hence μ_F is uniform.

(4) By Theorem 2.7, uniform observables imply $\mathrm{SQ}(F)$ is a subsingleton. Since F is inhabited (A1), $\mathrm{SQ}(F)$ is nonempty. A nonempty subsingleton is equivalent to $\mathbf{1}$. □

Corollary 8.2 (Framework Equivalence)**.** *Let $\mathcal{C} = (\mathbf{1}, \mathrm{id}, \mathcal{O}_{\text{dim}}, \mu_{\mathcal{C}})$ be the canonical one-state framework. If F satisfies* (A1)–(A5) *and there exists an observable-type equivalence $e : O_F \simeq \mathcal{O}_{\text{dim}}$, then*

$$\mathrm{SQ}(F) \simeq \mathrm{SQ}(\mathcal{C}) \simeq \mathbf{1} \qquad \textit{and} \qquad O_F \simeq O_{\mathcal{C}}.$$

Remark 8.3 (What the theorem does not claim). The theorem does *not* assert that the raw state space S_F is a singleton. It asserts that all internal states are observationally indistinguishable at the cost-extracted interface: multiplicity of S_F is gauge redundancy, not physical freedom.

The conditional uniqueness is now proved. Full inevitability—the claim that *any* $F \in \mathcal{Z}$ must satisfy (A3)–(A5) and therefore collapse—reduces to closing the bridges below. Bridge B2 (the composition law) is now closed (Theorem 9.3); the remaining bridges are well-posed intermediate results.

9 The Open Derivation Programme: Six Bridges

The full inevitability claim—"any $F \in \mathcal{Z}$ is forced into the RS structure"—requires deriving (A3)–(A5) from (A1)–(A2) alone. This section names the six structural bridges.

9.1 Bridge B1: self-similarity from zero parameters

Theorem 4.5 proves: *among* self-similar schemas, φ is forced. It does not prove: a zero-parameter ledger *must* be self-similar.

Open Problem 9.1 (Hierarchy Theorem)**.** Show that a discrete, zero-parameter ledger with at least two levels of event composition and uniform inter-level scaling necessarily satisfies a monic integer recurrence. This, combined with Theorem 4.5, forces φ.

The intermediate result "(A2) + hierarchical structure $\Rightarrow$ (A3)" is well-posed. Deriving hierarchical structure from (A2) alone remains open.

9.2 Bridge B2: the RCL from the factorization property (closed)

The RCL (1) was previously adopted as an axiom in (A4). We now derive it from a weaker, structurally necessary condition: the *factorization property*.

Definition 9.2 (Factorization Property)**.** A cost functional $J : \mathbb{R}_{>0} \to \mathbb{R}$ satisfies the *factorization property* if there exists a continuous function $\varphi : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ such that

$$J(xy) + J(x/y) = \varphi(J(x), J(y)) \qquad \text{for all } x, y > 0. \tag{5}$$

No specific form of φ is assumed.

The factorization property is a structural necessity: it says the total cost of a compound comparison (x, y) depends on x and y only through their individual costs $J(x)$ and $J(y)$. This is weaker than assuming the specific d'Alembert quadratic form. We show it suffices.

Theorem 9.3 (Factorization + Admissibility Forces the RCL)**.** *Let $J : \mathbb{R}_{>0} \to \mathbb{R}$ satisfy:*

1. *Reciprocal symmetry: $J(x) = J(x^{-1})$.*
2. *Unit normalization: $J(1) = 0$.*
3. *Strict convexity on $\mathbb{R}_{>0}$ (global: $J''(x) > 0$ for all $x > 0$).*
4. *Calibration: $(J \circ \exp)''(0) = 1$.*
5. *Continuity on $\mathbb{R}_{>0}$.*
6. *Factorization property (Theorem 9.2).*
7. *Zero-parameter posture (A2).*

Then $J = J_{\text{cost}}$ and $\varphi(a, b) = 2ab + 2a + 2b$ (the RCL).

Proof. Define $H(t) = J(e^t) + 1$, so $H(0) = 1$, H is even and continuous. The factorization property becomes

$$H(s+t) + H(s-t) = \Phi(H(s), H(t)) \tag{6}$$

for some continuous Φ.

Step 1: Boundary values. Setting $t = 0$: $2H(s) = \Phi(H(s), 1)$, hence $\Phi(a, 1) = 2a$. Setting $s = 0$: $2H(t) = \Phi(1, H(t))$ (using evenness), hence $\Phi(1, b) = 2b$.

Step 2: Triple identity forces affinity. Apply (6) with three different groupings of α, β, γ:

$$\begin{aligned} H(\alpha+\beta+\gamma) + H(\alpha+\beta-\gamma) &= \Phi(H(\alpha+\beta),\, H(\gamma)), \\ H(\alpha-\beta+\gamma) + H(\alpha-\beta-\gamma) &= \Phi(H(\alpha-\beta),\, H(\gamma)). \end{aligned}$$

Adding, and comparing with the groupings using α and $\beta \pm \gamma$, yields:

$$\Phi(H(\alpha{+}\beta),\, c) + \Phi(H(\alpha{-}\beta),\, c) = \Phi(a,\, H(\beta{+}\gamma)) + \Phi(a,\, H(\beta{-}\gamma))$$

where $a = H(\alpha)$, $c = H(\gamma)$. Setting $\gamma = 0$ gives $c = 1$, and $H(\beta{+}0){+}H(\beta{-}0) = \Phi(H(\beta), 1) = 2H(\beta)$. The identity becomes $2H(\alpha+\beta) + 2H(\alpha-\beta) = \Phi(a, p) + \Phi(a, q)$ where $p + q = \Phi(H(\beta), c)$. Since H is strictly convex, the pair (p, q) with fixed sum $p + q$ varies continuously over an interval as γ ranges over $[0, \infty)$. The constraint $\Phi(a, p) + \Phi(a, q) = \text{const}$ whenever $p + q = \text{const}$ forces $\Phi(a, \cdot)$ to be affine for each a:

$$\Phi(a, b) = \alpha(a)\, b + \beta(a).$$

(If f is continuous and $f(p) + f(q)$ depends only on $p + q$, then $f' = \text{const}$ on the relevant interval.)

Step 3: Determine Φ. From $\Phi(a, 1) = 2a$: $\alpha(a) + \beta(a) = 2a$. By symmetry $\Phi(a, b) = \Phi(b, a)$: $\alpha(a)b + \beta(a) = \alpha(b)a + \beta(b)$ for all a, b. Substituting $\beta(a) = 2a - \alpha(a)$:

$$\alpha(a)(b-1) - \alpha(b)(a-1) = 2(b-a).$$

Setting $b = 2$ gives $\alpha(a) = (4 - \alpha(2)) + a(\alpha(2) - 2)$, so α is affine: $\alpha(a) = (2 - c) + ca$ where $c = \alpha(2) - 2$. Write $c_2 = c + 2 = \alpha(2)$. Then:

$$\Phi(a, b) = c_2\, ab + (2 - c_2)(a + b - 1). \tag{7}$$

This is a one-parameter family indexed by $c_2 > 0$.

Step 4: Reduce to the d'Alembert equation. Define $p(t) = \frac{c_2}{2}\, H(t) + 1 - \frac{c_2}{2}$. Direct substitution verifies: $p(s+t) + p(s-t) = 2\, p(s)\, p(t)$ (the d'Alembert equation for p). By the Aczél classification (Theorem 5.3), $p(t) = \cosh(\lambda t)$ with $\lambda = \sqrt{c_2/2}$. Hence:

$$J(x) = \frac{1}{c_2}(x^\lambda + x^{-\lambda} - 2), \qquad \lambda = \sqrt{c_2/2}.$$

Step 5: Strict global convexity forces $\lambda \geq 1$. Compute:

$$J''(x) = \frac{\lambda}{c_2}\big[(\lambda - 1)\, x^{\lambda-2} + (\lambda + 1)\, x^{-\lambda-2}\big].$$

For $\lambda < 1$: the coefficient $(\lambda - 1) < 0$ and $x^{\lambda-2} \to 0$ slower than $x^{-\lambda-2} \to 0$ as $x \to \infty$, so $J''(x) < 0$ for sufficiently large x. Hence strict global convexity on $\mathbb{R}_{>0}$ requires $\lambda \geq 1$, i.e., $c_2 \geq 2$.

Step 6: Zero-parameter selection forces $\lambda = 1$. The exponent $\lambda = \sqrt{c_2/2}$ is a structural constant of the composition rule. For $\lambda > 1$, specifying λ requires encoding a non-trivial real number—a free parameter. Under the zero-parameter posture (A2), every structural constant must have $O(1)$ Kolmogorov complexity. The unique globally convex exponent specifiable with zero additional information is $\lambda = 1$ (the identity exponent: $x^1 = x$), giving $c_2 = 2$.

Step 7: Conclude. $\lambda = 1$ gives $J(x) = \frac{1}{2}(x + x^{-1}) - 1 = J_{\text{cost}}(x)$ and $\Phi(a, b) = 2ab$, i.e., $\varphi(J(x), J(y)) = 2J(x)J(y) + 2J(x) + 2J(y)$: the RCL. □

Remark 9.4 (What this closure means). Bridge B2 is now closed. The RCL is no longer an adopted axiom—it is *derived* from conditions (1)–(7), where (1)–(5) are the original admissibility conditions, (6) is the factorization property (a structural necessity: compound costs factor through individual costs), and (7) is the zero-parameter posture already in (A2). The factorization property replaces the specific d'Alembert form as the input; the d'Alembert form emerges as the output.

Remark 9.5 (Status of the proof). Steps 1–5 are rigorous mathematics with complete proofs. Step 6 uses the Kolmogorov-complexity / AIT framework that the paper already employs for the zero-parameter definition (A2). In particular, the argument "$\lambda = 1$ is the unique parameter-free exponent" is of the same logical type as "the state space admits an algorithmic specification." If one accepts (A2), one should accept Step 6.

9.3 Bridge B3: the ratio-extraction interface

(A5) posits $r : S_F \to \mathbb{R}_{>0}$. This bundles three sub-claims:

1. *Dimensionlessness:* all observables are ratios (no dimensionful constants in a zero-parameter framework).
2. *One-dimensionality:* a single real per state (requires the ledger to have one independent conserved quantity).
3. *Positivity:* the ratio lives on $\mathbb{R}_{>0}$.

9.4 Bridge B4: the ground-state restriction

The hypothesis $J(r(s)) = 0$ for all s is not automatic. By reciprocal symmetry, $r(S_F)$ is inversion-closed. Under a *structural determinacy* (SD) hypothesis—that each ratio is uniquely determined by a computable, $\mathbb{Z}/2\mathbb{Z}$-invariant predicate—any $x \neq 1$ requires at least 1 bit to select from $\{x, x^{-1}\}$, contradicting zero parameters. On finite graphs, unique minimizers within flux sectors are established in [7] via uniform convexity of $\sum_E J(x_v/x_w)$ in log variables. Deriving (SD) from (A2) alone in the abstract setting is open.

Example 9.6 (Concrete falsifier for automatic ground-state)**.** Let $S_F = \{s_0\}$, $r(s_0) = 2$. Then $J_{\text{cost}}(r(s_0)) = J_{\text{cost}}(2) = \frac{1}{4} \neq 0$. This one-state framework satisfies (A1), (A2), (A4), and vacuously (A5-i), but violates the ground-state restriction.

9.5 Bridge B5: the constructive prediction map

Theorem 8.1 proves μ_F is constant but does not identify the constant. A prediction map $\mathcal{P} : (J_{\text{cost}}, \varphi) \to \mathcal{O}_{\text{dim}}$ would need to: (i) take J_{cost} and φ as sole inputs, (ii) produce specific dimensionless quantities, (iii) have $O(1)$ description length, and (iv) be unique. Existence (i)–(iii) can be established by formalizing the RS ledger algorithm; the concrete construction of the recognition operator and

its spectral structure are developed in [9], and the connection to measurement probabilities in [10]. Uniqueness (iv) and value identification remain open.

9.6 Bridge B6: the observable-type equivalence

Theorem 8.2 requires $e : O_F \simeq \mathcal{O}_{\mathrm{dim}}$ as input. After quotient collapse, $\mathrm{im}(\mu_F)$ is a singleton. Any two singleton types are abstractly isomorphic, so the type-level equivalence is trivially constructible. The recognition-geometric formalization of quotients by observational indistinguishability is developed in [8]. Non-trivial content (value identification) reduces to B5.

10 Falsifiers

The theorem is a mathematical statement that cannot be refuted within its hypotheses. However, the *inevitability framing* can be challenged by showing that a hypothesis is too strong, too narrow, or physically unwarranted. Each of the following would constitute a substantive refutation:

1. **Alternative zero-parameter framework.** Construct $F \in \mathcal{Z}$ that derives observables and does *not* collapse on the quotient (non-uniform μ_F without free parameters).

2. **Non-φ self-similarity.** Provide a parameter-free self-similar scaling whose preferred scale is not φ, using a notion of self-similarity that retains the intended physical meaning.

3. **Continuous zero-parameter physics.** Exhibit a theory with no adjustable dimensionless knobs whose state space is genuinely uncountable, and argue that continuity does not constitute hidden parameter freedom.

   Worked example. Consider pure general relativity (no matter, no cosmological constant). The Einstein–Hilbert action $S = \int R\sqrt{|g|}\, d^4x$ has no adjustable coupling constants. Yet the space of solutions (Lorentzian metrics modulo diffeomorphisms) is uncountable. Under our definition of zero parameters (Theorem 2.3), the state space must admit an algorithmic specification covering every state. The moduli space of metrics does not: specifying an arbitrary smooth metric requires infinitely many real parameters (Fourier coefficients, initial data on a Cauchy surface, etc.). Therefore pure GR does *not* satisfy (A2) under the algorithmic-specification definition, and the exclusivity theorem does not apply to it. A compelling refutation would either: (a) find a finitary parametrization of the GR solution space (contradicting well-known results), or (b) argue that the algorithmic-specification definition is too restrictive for "zero parameters" as physically intended.

4. **Conservation without double-entry.** Produce a discrete conserved dynamics where conservation is representable without a balanced accounting structure.

5. **Pathological d'Alembert solution.** Construct a non-standard solution to (3) satisfying the raw identity but violating regularity. This would not refute Theorem 6.1 (regularity is explicit), but would show that regularity is a load-bearing hypothesis.

11 Conclusion

We have proved that any physics framework satisfying the five admissibility conditions has a unique observational state on the quotient: $\mathrm{SQ}(F) \simeq \mathbf{1}$. The preferred scale is $\varphi = (1+\sqrt{5})/2$ and the cost functional is $J_{\mathrm{cost}}(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Two of the five assumptions remain adopted axioms (self-similarity and cost-minimum semantics). The third previously open assumption—the composition law—is now derived: Theorem 9.3 shows that the factorization property, combined with strict convexity and the zero-parameter posture, uniquely forces the d'Alembert/RCL form. This closes Bridge B2, which was previously the deepest open problem.

Of the remaining five bridges, B1 (self-similarity from discreteness), B3 (ratio interface), and B4 (ground-state restriction) are well-posed intermediate results with clear proof strategies. B5 (prediction map uniqueness) and B6 (value identification) do not affect the conditional theorem.

The programme has advanced from "six open bridges" to "two adopted axioms plus three well-posed intermediate results." The composition law is no longer assumed—it is a theorem.

Connection to the Recognition Science programme. Once the conditional quotient collapse is established, the unique J_{cost} and φ directly determine the downstream structure: the discrete ledger and its conservation laws [6], the 8-tick carrier (period 2^D with $D = 3$, forced by φ-scaling on the 3-cube Q_3) with its recognition-operator dynamics [9], the gauge quotient ($\mathrm{Aut}(Q_3) \cong S_4 \times \mathbb{Z}_2$ forcing $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$) [8], the mass law ($m = A_s \cdot \varphi^{r-8+\mathrm{gap}(Z)}$) [2], and the Born rule from the cost-plus-geometry input [10]. The quotient collapse proved here is the foundational layer beneath all of these: it establishes that the structural inputs (J_{cost}, φ) are not chosen but forced, so everything downstream is forced as well.

A Formal Verification

All proved results are formally verified in Lean 4 (v4.27.0-rc1, Mathlib4). The complete formalization lives at:

https://github.com/jonwashburn/recognition-science

All files relevant to this paper are under `RecognitionScience/Verification/Exclusivity/`. The codebase uses one external axiom: `aczel_representation_axiom` (Theorem 5.3), encoding the Aczél classification theorem; all other steps are fully proved.

File map

Paper result	Lean theorem	File (prefix `Exclusivity/`)
Core definitions		
Theorem 2.1	`PhysicsFramework`	`Framework.lean`
Theorems 2.4 and 2.6	`ObsEquiv`, `StateQuotient`	`Framework.lean`
Theorem 2.7	`quotient_subsingleton`	`Framework.lean`
Main theorem components		
Theorem 4.5 (φ forcing)	`selfSim_fibonacci`	`ModelIndependent.lean`
Theorem 6.1 (cost uniqueness T5)	`washburn_zlatanovic_cost`	`CostFunctionalEquation.lean`
Theorem 7.3 (derived uniformity)	`derived_uniform_observables`	`ModelIndependent.lean`
Theorem 8.1 (Theorem 7.1, quotient collapse)	`derived_model_independent`	`ModelIndependent.lean`
Theorem 8.2 (quotient $\simeq \mathbf{1}$)	`quotient_equiv_unit`	`ModelIndependent.lean`
Bridge B1 (proved, zero `sorry`)		
Section 9.1	`hierarchy_forces_fibonacci`	`HierarchyTheorem.lean`
Bridge B3 (ratio existence proved)		
Section 9.3	`bridge_B3_single_channel`	`DimensionlessRatio.lean`
Bridge B4 (corollary proved; core sorry)		
Section 9.4 corollary	`ground_state_implies_constant`	`GroundStateRestriction.lean`
Section 9.4 falsifier	`example : Cost.Jcost 2 ≠ 0`	`GroundStateRestriction.lean`
Bridge B6 (proved, zero `sorry`)		
Section 9.6	`bridge_B6_quotient_equiv`	`ObservableTypeEquiv.lean`
Bridge B2 scaffold (Open Problem B is `sorry`)		
Boundary conditions	`composition_rule_f0`	`RGDerivation.lean`
Boundary conditions	`composition_rule_f_at_zero`	`RGDerivation.lean`
Theorem 9.3 (B2 closure)	`composition_rule_classification`	`RGDerivation.lean`
Bridge B5 scaffold (Open Problems 2–3 are `sorry`)		
OP1 (existence)	`bridge_B5_prediction_map_exists`	`PredictionMap.lean`
OP2 (uniqueness)	`prediction_map_unique`	`PredictionMap.lean`
Master certificate		
All of Thm. 7.1	`ExclusivityV2Cert.verify`	`ModelIndependentCert.lean`

Epistemic status summary

Result	Sorry?	Note
Framework, quotient, uniformity lemmas	0	Fully proved
Self-similarity forces φ	0	Fully proved
T5 cost uniqueness	0	Conditional on Aczél axiom
Main Theorem 7.1	0	Conditional on Aczél axiom
B1 hierarchy recurrence	0	Fully proved
B6 quotient $\simeq \mathbf{1}$	0	Fully proved
B3 ratio extraction	1	AIT dimensionless argument open
B4 ground-state core	1	SD derivation from (A2) open
B2 composition classification	1	**Closed in paper** (Thm. 9.3); Lean scaffold pending
B5 prediction map uniqueness	1	Open Problem 2

To check out and build:

```
git clone https://github.com/jonwashburn/recognition-science
cd recognition-science
lake exe cache get   # downloads prebuilt Mathlib (~2 GB)
lake build RecognitionScience.Verification.Exclusivity.ModelIndependentCert
```

B Counter-Example: A Zero-Parameter Framework Without the RCL

We exhibit a framework that satisfies (A1) and (A2) but violates the RCL, producing non-collapsing observables. This shows that the composition law is load-bearing: without it (or something equivalent), quotient collapse fails.

Example B.1 (Log-cost framework)**.** Define F with $S_F = \mathbb{R}_{>0}$ (ratios), $T_F = \mathrm{id}$, $O_F = \mathbb{R}$, and

$$\mu_F(x) = \log x, \qquad J_{\log}(x) = |\log x|.$$

Then:

1. S_F admits an algorithmic specification (enumerate dyadic rationals; decode via standard binary representation). Hence $F \in \mathcal{Z}$.
2. $J_{\log}$ satisfies reciprocal symmetry: $|\log x| = |\log x^{-1}|$.
3. $J_{\log}(1) = 0$ (unit normalization).
4. $J_{\log}$ is *not* strictly convex on $\mathbb{R}_{>0}$ (it is V-shaped, not smoothly curved).
5. The composition is additive: $J_{\log}(xy) + J_{\log}(x/y) = |\log x + \log y| + |\log x - \log y|$, which does *not* equal $2J_{\log}(x)J_{\log}(y) + 2J_{\log}(x) + 2J_{\log}(y)$ in general. Hence the RCL fails.
6. Observables are non-uniform: $\mu_F(2) = \log 2 \neq 0 = \mu_F(1)$. The quotient does *not* collapse.

Remark B.2. This framework satisfies (A1), (A2), and partial versions of (A4) (symmetry + normalization), but fails strict convexity and the composition law. It demonstrates that the RCL and convexity are not automatic consequences of the zero-parameter posture; they are *additional* structural requirements that the ledger must impose. Without them, a zero-parameter framework can produce non-uniform observables and the quotient survives with more than one equivalence class.

Remark B.3 (What the counter-example shows about Bridge B2). The log-cost framework is ruled out by any of: (a) strict convexity, (b) smooth calibration, or (c) the RCL. It is *not* ruled out by (A1)–(A2) alone. This is precisely why Bridge B2 matters: the RCL must be *derived* from ledger structure (not from the zero-parameter definition by itself) for the inevitability claim to close.

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